

Sequence of R scripts for Logistic Regression 2 August 2026

Preprocess Chl by MARSS-DLM and merge with daily covariates:

“Read+DLM-by-MARSS_ChL_Merged_Dailies_MultiYear_2026-01-19.R”

The result for daily manual chlorophyll, sqrt transformed, contains:

```
# Save results for spline fit of Langevin
# Include a list of input data: xoriginal is untransformed, Xvar0 is sqrt(xoriginal),
# mean and sd are mu.Xvar0 and sd.Xvar0, XZ is Z-score of Xvar0
# Xtrans.lst = list(xoriginal,Xvar0,mu.Xvar0,sd.Xvar0,XZ)
save(dat0c,Xtrans.lst,                                     # input data & transform
     mat1,xvar0,xvar1,Tindx,TT,DT,                         # inputs to MARSS
     b0,b1,b0.se,b1.se,stdlevel,yhat,sm.y,sm.yse,          # outputs of MARSS
     file=Fname)
```

Outputs:

File name (Fname) is "DLM-MARSS_result_ChL_PeterSqueal.Rdata"

Plot Fig. S1

Check Markov property and stationarity

“Diagnostics_ADF_stationarity_ARIMA_Markov_2023-09-05.R”

Reconstruct Langevin model by spline and compute probabilities of crossing repellor

“Read+Spline_ChL_MARSS-DLM_result_15March2026.R”

This R script fits the Langevin model to daily smoothed prediction of sqrt(Chl), integrates the EPF, takes the derivative to find equilibria, and identifies the repellor. On each day the prediction and its s.e. are calculated from the spline model. The distance of each prediction from the repellor and the s.e. are combined to calculate the Student-t statistic for distance from repellor on each day, and the probability that the system is in the low-chl state is estimated in R.

Outputs:

File name is ‘Spline_MARSSdlm_predix-tprobs_squeal1.Rdata’ contains data frame ‘dat4reg’ with the Student t results, probabilities for log(oddsratio) for the low stable state of chlorophyll and covariates.

Plots of Fig. 2, Fig. 3

The next step is to merge this data frame with the data for LMB and pelagic minnows.

Merge spline result with LMB, pelagic minnow and zooplankton biomass

Run the script “Combine_Raft+Zoop+Fish_08-11_squeal1_v1_2026-02-10.R”. The script generates an updated dataframe ‘all4reg’ containing all covariates for logistic regression analyses – the response variates (t-statistics and probabilities), the covariates measured by sonde, daily zooplankton biomass and mean *Daphnia* length, littoral minnows, pelagic minnow index, and LMB.

R line to load the merged result:

```
load(file='CombineCovarSplineSqueal1.Rdata')
```

Plots and Regressions

Plot Fig. 1: “Plot_FoodWeb_TimeSeries_2008-2011.R”

Regress the log(odds ratio) for Chlorophyll versus food chain elements to calculate the statistics for Fig. 5 and Fig. 6 by running

```
“ORR_TransformFoodChain_V2spline_LogOdds_19Feb2026.R”
```

Plot Fig. 4 using the regression of log(odds ratio) for the low stable state of chlorophyll versus adult largemouth bass population (LMB) using the bootstrapped standard deviation of the unstable threshold between stable states of chlorophyll (see below):

```
“Plots_Final_LMB_model_2026-07-26.R”
```

Construct Fig. 5 in PowerPoint using statistics from the regressions printed by

```
“ORR_TransformFoodChain_V2spline_LogOdds_19Feb2026.R”
```

Bootstrap Estimates of Standard Deviations of Chlorophyll Equilibria

Errors of the chlorophyll equilibria were estimated by bootstrapping the estimates of the Langevin model. Bootstraps were constructed by randomly resequencing the residuals of the spline model for the drift function. The bootstrap was run for 1000 cycles in about 2 hours on a Dell Precision 3240 desktop computer using 20 cores by

"\Bootstrap+Spline_ChL_after-MARSS-DLM_17March2026.R"

The number of bootstrap cycles is set in line 175 and the output file name is set in line 170. The default number of cycles is 10.

Bootstrap result were summarized and plotted by

"After-Boot_R-Chl-0811_Mean+SD_2026-07-20.R"

Plots of Fig. S2, Fig. S3

Standard deviations of equilibria were computed using the bootstrap cycles that had 3 equilibria. The other cycles had an unstable point just above zero (as occurs in the logistic equation for population growth and many other ecological models) and/or weakly-defined equilibria at high concentrations of chlorophyll in addition to the 3 equilibria supported by many data points. When analyzing data manually we ignore equilibria near zero. At high chlorophyll concentrations data are sparse and apparent equilibria may be spurious. We adjust the knots of the spline to smooth over high equilibria weakly supported by sparse data.